Patient Flow Simulator — Decision Report

Type 1 Emergency Department

Executive summary

- Modelled demand: 322 attendances/day (1,934 patients over the simulated period).
- 4-hour standard: 82% [MEETS the >=78% target]; 95% CI 77-82% over 24 runs; 347 breaches.
- Mean time in department: 2.9 h (95% CI 2.9-3.6 h); 95th percentile 9.4 h.
- Admitted: 18% of attendances.

Bottleneck diagnosis

Biggest delay: Wait for MH bed — patients wait 224 min on average (220 patient-hours lost over the run).

Recommendation: mh inpatient bed is busy (69% utilisation) and likely drives this queue; trial extra capacity in Scenario comparison.

UEC Plan 2025/26 scorecard

4-hour standard: 82% (target >=78%)	**GREEN**
12-hour waits: 3% (target <10%)	**GREEN**
Ambulance handover <45 min: 94% (target >=95%)	**AMBER**
Mental health >24h in ED: 17% (target <5%)	**RED**
Children's 4-hour standard: 88% (target >=80%)	**GREEN**

What-if scenarios (ranked)

Scenario	4-hr perf.	vs baseline	MH >24h	Mean LOS
Baseline	81%	—	15%	3.1 h
+3 mh inpatient bed	81%	**+0 pts**	5% (-11)	3.0 h
+8 inpatient bed	81%	**+0 pts**	15% (+0)	3.1 h
Winter +15% demand	77%	**-4 pts**	38% (+23)	3.6 h

Patient Flow Simulator

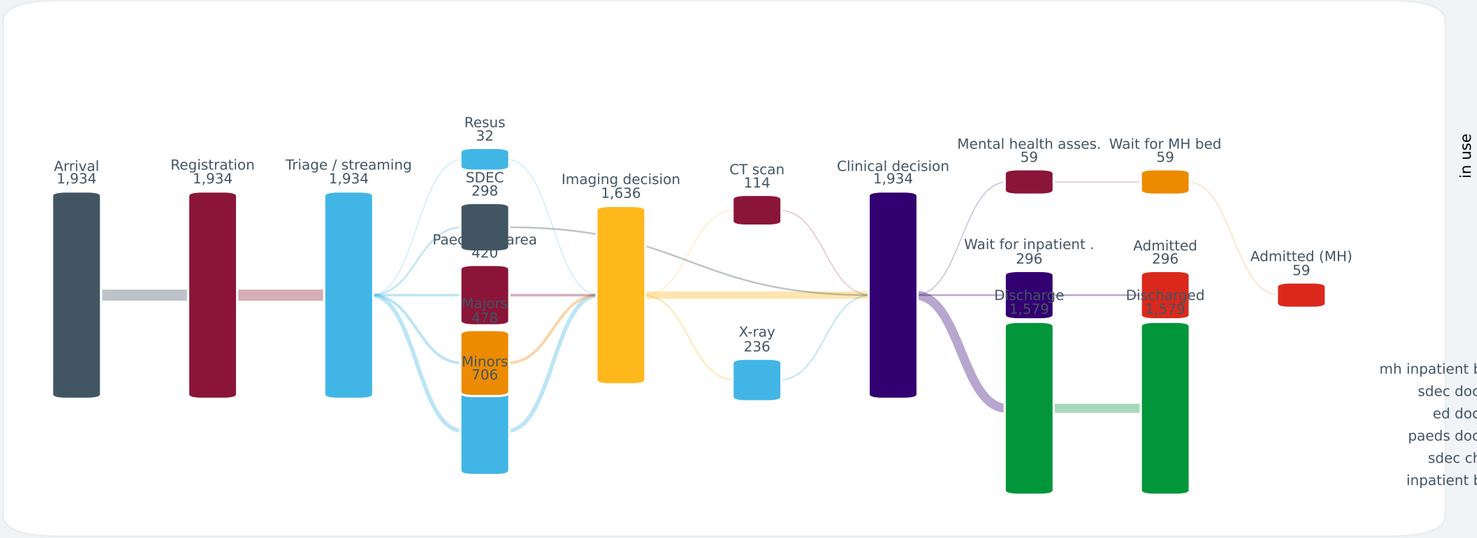
Discrete-event simulation for emergency & urgent care planning

Type 1 Emergency Department

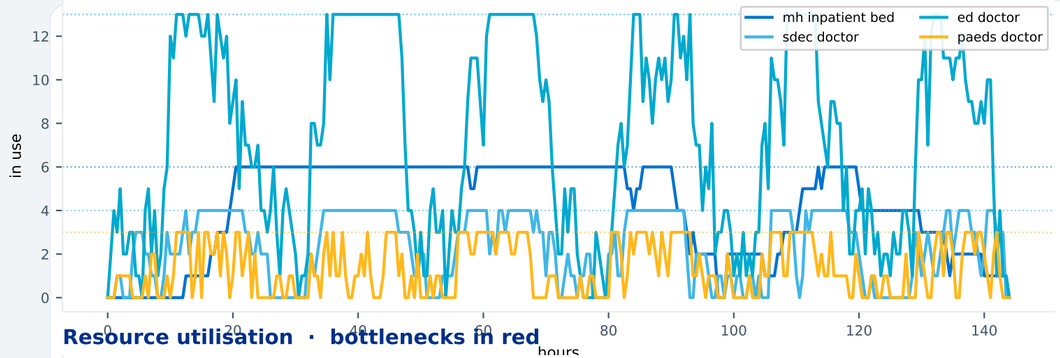
322 attendances/day · 6-day run · 24 Monte-Carlo replications

4-HOUR PERFORMANCE	MEAN TIME IN DEPT	95TH PERCENTILE LOS	ADMITTED	DAILY ATTENDANCES
82%	2.9h	9.4h	18%	322
95% CI 77-82%	95% CI 2.9-3.6h	longest typical wait	8.5h mean LOS	1,934 simulated

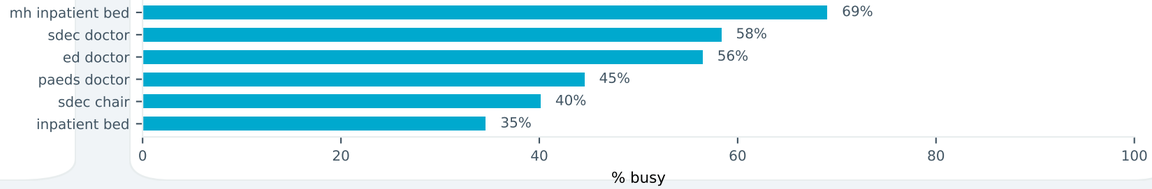
Patient flow · arrival to disposition



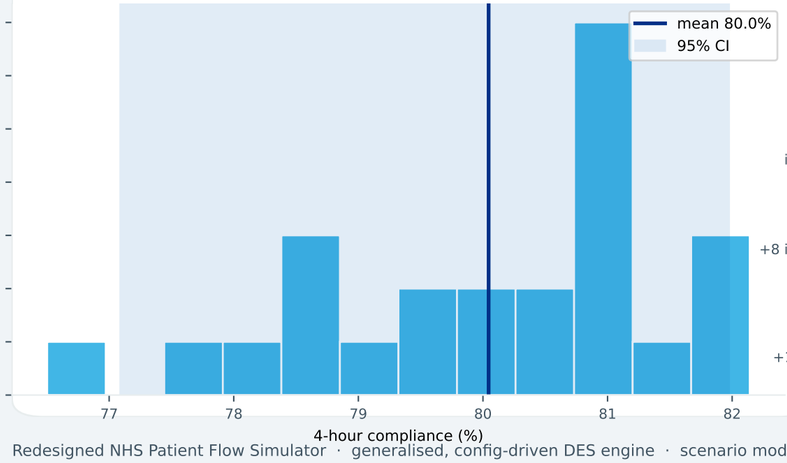
Resource occupancy vs capacity (dotted)



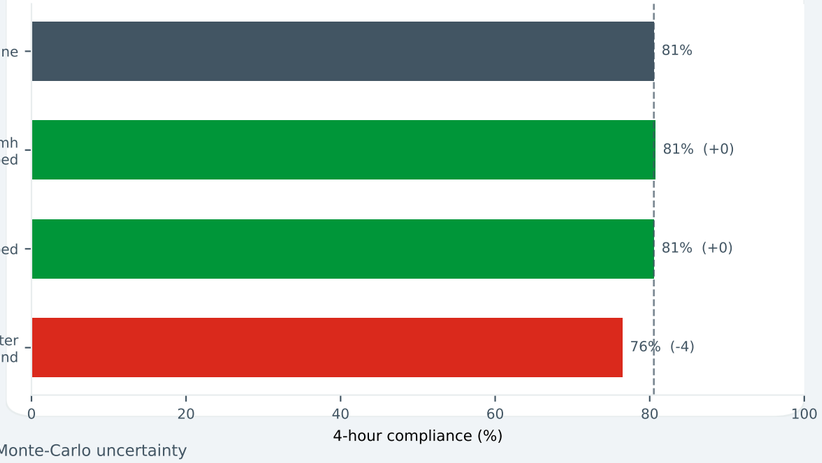
Resource utilisation · bottlenecks in red



Monte-Carlo: 4-hour performance variability



Scenario comparison · 4-hour performance



Where patients wait · mean queue time

STAGE	WAIT	SEEN
Wait for MH bed	**224m**	59
SDEC	**34m**	298
Mental health assessment	**13m**	59
Majors	**10m**	478
Minors	**10m**	706
Paediatric area	**6m**	420

For demonstration — synthetic data